CERTIFICATE OF ANALYSIS

Work Order	: **ES2604846**	Page	: 1 of 5
Client	: **AECOM AUSTRALIA PTY LTD**	Laboratory	: Environmental Division Sydney
Contact	: Caitlin Friedman	Contact	: Loren Schiavon
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Telephone	: ----	Telephone	: +61 2 8784 8555
Project	: 60704593	Date Samples Received	: 11-Feb-2026 10:00
Order number	: 60704593-7	Date Analysis Commenced	: 19-Feb-2026
C-O-C number	: ----	Issue Date	: 20-Feb-2026 17:17
Sampler	: Alex Liang		
Site	: ccc-Bateau Bay		
Quote number	: EN/004/23		
No. of samples received	: 15		
No. of samples analysed	: 15		

Accreditation No. 825
Accredited for compliance with
ISO/IEC 17025 - Testing

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted, unless the sampling was conducted by ALS. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QA/QC Compliance Assessment to assist with Quality Review and Sample Receipt Notification.

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is carried out in compliance with procedures specified in 21 CFR Part 11.

Signatories	*Position*	*Accreditation Category*
Ankit Joshi	Senior Chemist - Inorganics	Sydney Inorganics, Smithfield, NSW
Layla Hafner	Acid Sulphate Soils - Chemist	Brisbane Acid Sulphate Soils, Stafford, QLD

General Comments

The analytical procedures used by ALS have been developed from established internationally recognised procedures such as those published by the USEPA, APHA, AS and NEPM. In house developed procedures are fully validated and are often at the client request.

Where moisture determination has been performed, results are reported on a dry weight basis.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

When sampling time information is not provided by the client, sampling dates are shown without a time component. In these instances, the time component has been assumed by the laboratory for processing purposes.

Where a result is required to meet compliance limits the associated uncertainty must be considered. Refer to the ALS Contract for details.

Key : CAS Number = CAS registry number from database maintained by Chemical Abstracts Services. The Chemical Abstracts Service is a division of the American Chemical Society.
LOR = Limit of reporting
^ = This result is computed from individual analyte detections at or above the level of reporting
ø = ALS is not NATA accredited for these tests.
~ = Indicates an estimated value.

- ED045G: LOR raised for Chloride on sample 7due to sample matrix
- ASS: EA037 (Rapid Field and F(ox) screening): pH F(ox) Reaction Rate: 1 - Slight; 2 - Moderate; 3 - Strong; 4 - Extreme
- EA037 ASS Field Screening: NATA accreditation does not cover performance of this service.
- ED045G: The presence of Thiocyanate, Thiosulfate and Sulfite can positively contribute to the chloride result, thereby may bias results higher than expected. Results should be scrutinised accordingly.

Analytical Results

Sub-Matrix: SOIL (Matrix: SOIL)			Sample ID	BH01-0-0.6	BH02-0-1.8	BH02-1.8-2.5	BH03-0-0.9	BH03-0.9-2.0
			Sampling date / time	11-Feb-2026 00:00	11-Feb-2026 00:00	11-Feb-2026 00:00	10-Feb-2026 00:00	10-Feb-2026 00:00
Compound	CAS Number	LOR	Unit	ES2604846-001	ES2604846-002	ES2604846-003	ES2604846-004	ES2604846-005
				Result	Result	Result	Result	Result
EA002: pH 1:5 (Soils)								
pH Value	----	0.1	pH Unit	**7.9**	**8.5**	----	----	**5.6**
EA010: Conductivity (1:5)								
Electrical Conductivity @ 25°C	----	1	µS/cm	**53**	**81**	----	----	**307**
EA037: Ass Field Screening Analysis								
ø pH (F)	----	0.1	pH Unit	**8.2**	**8.4**	**8.4**	**6.3**	**5.9**
ø pH (Fox)	----	0.1	pH Unit	**5.4**	**5.8**	**3.5**	**2.8**	**3.5**
ø Reaction Rate	----	1	-	**3**	**3**	**2**	**3**	**2**
EA055: Moisture Content (Dried @ 105-110°C)								
Moisture Content	----	1.0	%	**13.6**	**15.8**	----	----	**31.1**
EA080: Resistivity								
Resistivity at 25°C	----	1	ohm cm	**18900**	**12300**	----	----	**3260**
ED040S : Soluble Sulfate by ICPAES								
Sulfate as SO4 2-	14808-79-8	10	mg/kg	<10	**20**	----	----	**110**
ED045G: Chloride by Discrete Analyser								
Chloride	16887-00-6	10	mg/kg	**40**	<10	----	----	**620**

Analytical Results

Sub-Matrix: SOIL (Matrix: SOIL)			Sample ID	BH04-0-0.5	BH04-0.5-1.0	BH04-1.0-2.0	BH05-0-0.5	BH05-0.3-1.0
			Sampling date / time	10-Feb-2026 00:00	10-Feb-2026 00:00	10-Feb-2026 00:00	10-Feb-2026 00:00	10-Feb-2026 00:00
Compound	CAS Number	LOR	Unit	ES2604846-006	ES2604846-007	ES2604846-008	ES2604846-009	ES2604846-010
				Result	Result	Result	Result	Result
EA002: pH 1:5 (Soils)								
pH Value	----	0.1	pH Unit	----	7.3	----	----	8.6
EA010: Conductivity (1:5)								
Electrical Conductivity @ 25°C	----	1	µS/cm	----	32	----	----	80
EA037: Ass Field Screening Analysis								
ø pH (F)	----	0.1	pH Unit	7.4	7.2	6.9	8.4	8.3
ø pH (Fox)	----	0.1	pH Unit	5.1	4.4	4.0	4.8	5.0
ø Reaction Rate	----	1	-	3	3	3	3	3
EA055: Moisture Content (Dried @ 105-110°C)								
Moisture Content	----	1.0	%	----	22.8	----	----	10.6
EA080: Resistivity								
Resistivity at 25°C	----	1	ohm cm	----	31200	----	----	12500
ED040S : Soluble Sulfate by ICPAES								
Sulfate as SO4 2-	14808-79-8	10	mg/kg	----	10	----	----	30
ED045G: Chloride by Discrete Analyser								
Chloride	16887-00-6	10	mg/kg	----	<50	----	----	70

Analytical Results

Sub-Matrix: SOIL (Matrix: SOIL)			Sample ID	BH05-1.0-2.0	BH06-0-0.5	BH06-0.5-0.6	BH06-0.6-1.0	BH06-1.0-2.0
			Sampling date / time	10-Feb-2026 00:00	10-Feb-2026 00:00	10-Feb-2026 00:00	10-Feb-2026 00:00	10-Feb-2026 00:00
Compound	CAS Number	LOR	Unit	ES2604846-011	ES2604846-012	ES2604846-013	ES2604846-014	ES2604846-015
				Result	Result	Result	Result	Result
EA002: pH 1:5 (Soils)								
pH Value	----	0.1	pH Unit	----	----	----	7.4	----
EA010: Conductivity (1:5)								
Electrical Conductivity @ 25°C	----	1	µS/cm	----	----	----	79	----
EA037: Ass Field Screening Analysis								
ø pH (F)	----	0.1	pH Unit	7.8	8.4	8.3	7.8	7.7
ø pH (Fox)	----	0.1	pH Unit	3.5	6.0	5.3	2.9	3.4
ø Reaction Rate	----	1	-	3	3	3	3	2
EA055: Moisture Content (Dried @ 105-110°C)								
Moisture Content	----	1.0	%	----	----	----	13.6	----
EA080: Resistivity								
Resistivity at 25°C	----	1	ohm cm	----	----	----	12700	----
ED040S : Soluble Sulfate by ICPAES								
Sulfate as SO4 2-	14808-79-8	10	mg/kg	----	----	----	60	----
ED045G: Chloride by Discrete Analyser								
Chloride	16887-00-6	10	mg/kg	----	----	----	80	----

Inter-Laboratory Testing

Analysis conducted by ALS Brisbane, NATA accreditation no. 825, site no. 818 (Chemistry / Biology).

(SOIL) EA037: Ass Field Screening Analysis